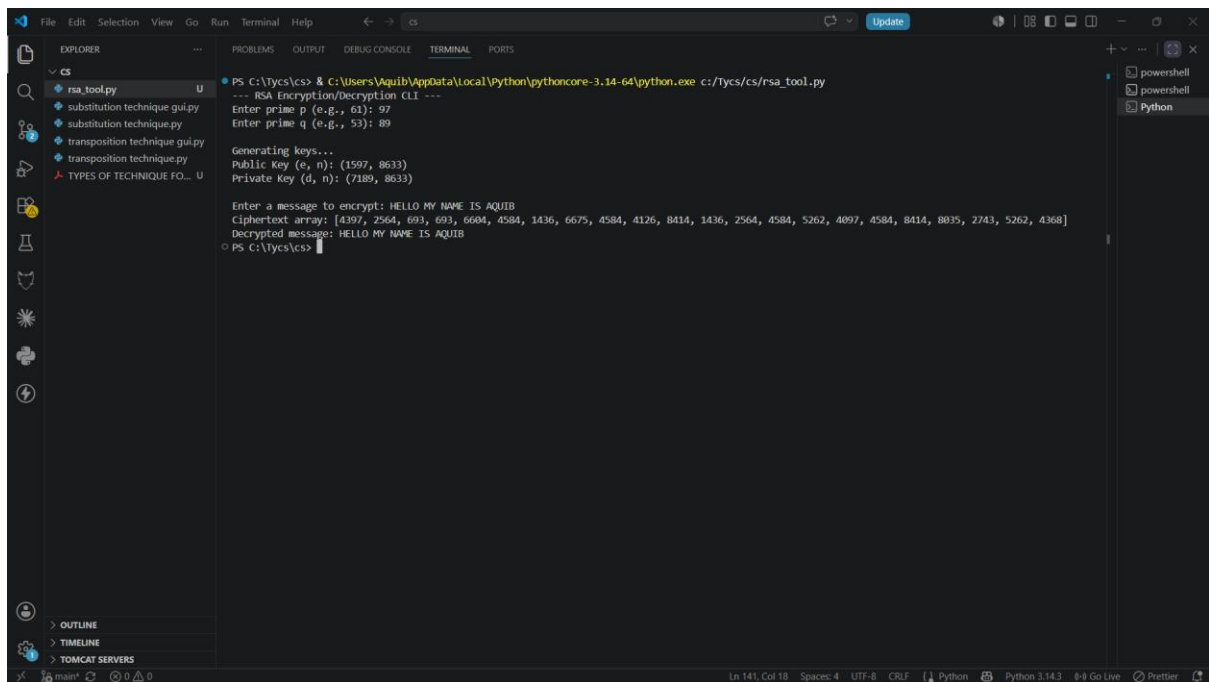


AIM: Implement the RSA algorithm for public-key encryption and decryption, and explore its properties and security considerations.

OUTPUT: CLI OF RSA ENCRYPTION AND DECRYPTION



```
PS C:\Tycs\cs> & C:\Users\Aqib\AppData\Local\Python\pythoncore-3.14-64\python.exe c:\Tycs\cs\rsa_tool.py
--- RSA Encryption/Decryption CLI ---
Enter prime p (e.g., 61): 97
Enter prime q (e.g., 53): 89

Generating keys...
Public Key (e, n): (1597, 8633)
Private Key (d, n): (7189, 8633)

Enter a message to encrypt: HELLO MY NAME IS AQUIB
Ciphertext array: [4397, 2564, 693, 6684, 4584, 1436, 6675, 4584, 4126, 8414, 1436, 2564, 4584, 5262, 4097, 4584, 8414, 8035, 2743, 5262, 4368]
Decrypted message: HELLO MY NAME IS AQUIB
PS C:\Tycs\cs>
```

OUTPUT: GUI OF RSA ENCRYPTION AND DECRYPTION

